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CytoScope

The Automated Semi-Robotic Cell Imaging System

Literature Review & Market Analysis (Industry)

Pros
Compact
High functionality (cell count, cytotoxicity, morphological features)
Software automates cell count, feature extraction
Easy installation

Cons
Expensive ($10,000-$50,000)
Parts are not easily able to be replaced, serviced
Low-throughput (one plate)

CellCyte X

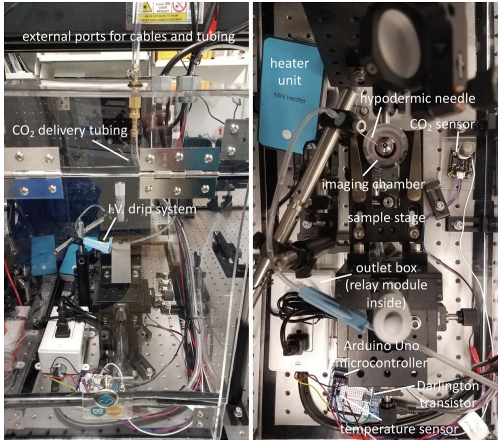
zenCELL Owl

Commercially available in-incubator microscopes are expensive and hard to maintain.

Literature Review & Market Analysis (Lab)

Pros
Low-cost
Modular parts (assembled from 3D printed / off the shelf)
Open-source software
High functionality (cell count, z-stack images)

Cons
Difficult assembly and removal
Setup involves modifying whole incubator
Complex wiring, microscope, and lighting
Low-throughput



Picroscope (Ly et al.) **Imaging incubator (Rossi et al.)**

Current publications' in-incubator microscopes require major incubator modification, and therefore assembly is difficult.

Our System - Goals

Goals
Low-cost (Under $2000)
Modular parts (3D printed/off the shelf)
High capacity (multiple cell plates)
Open-source software
Compact (fit within existing incubators)
Simple assembly/installation

Our system should be *easy to use* and have a *high throughput*.

Solution Development

Technical Specifications

Materials

Prototyping

Testing

Technical Specifications

Size	• Must be able to fit inside of an incubator • Can hold six 6-well plates • Under 10kg • Quick removal - 1 min
Compatible with Cell Culture Standard Protocols	• Cells remain immobile inside incubator • Cells are not exposed to continuous light • No contact with the cell plate
User Interface	• No assembly required • Images viewable on lab computer • Remote control of imaging parameters (focus, wells imaged) • Remote start at a push of a button in 2min
Maintenance and Calibration	• No daily upkeep • Can be sterilized with 70% ethanol in less than 5 mins • Repair parts easily sourced and made • Calibration takes less than one minute

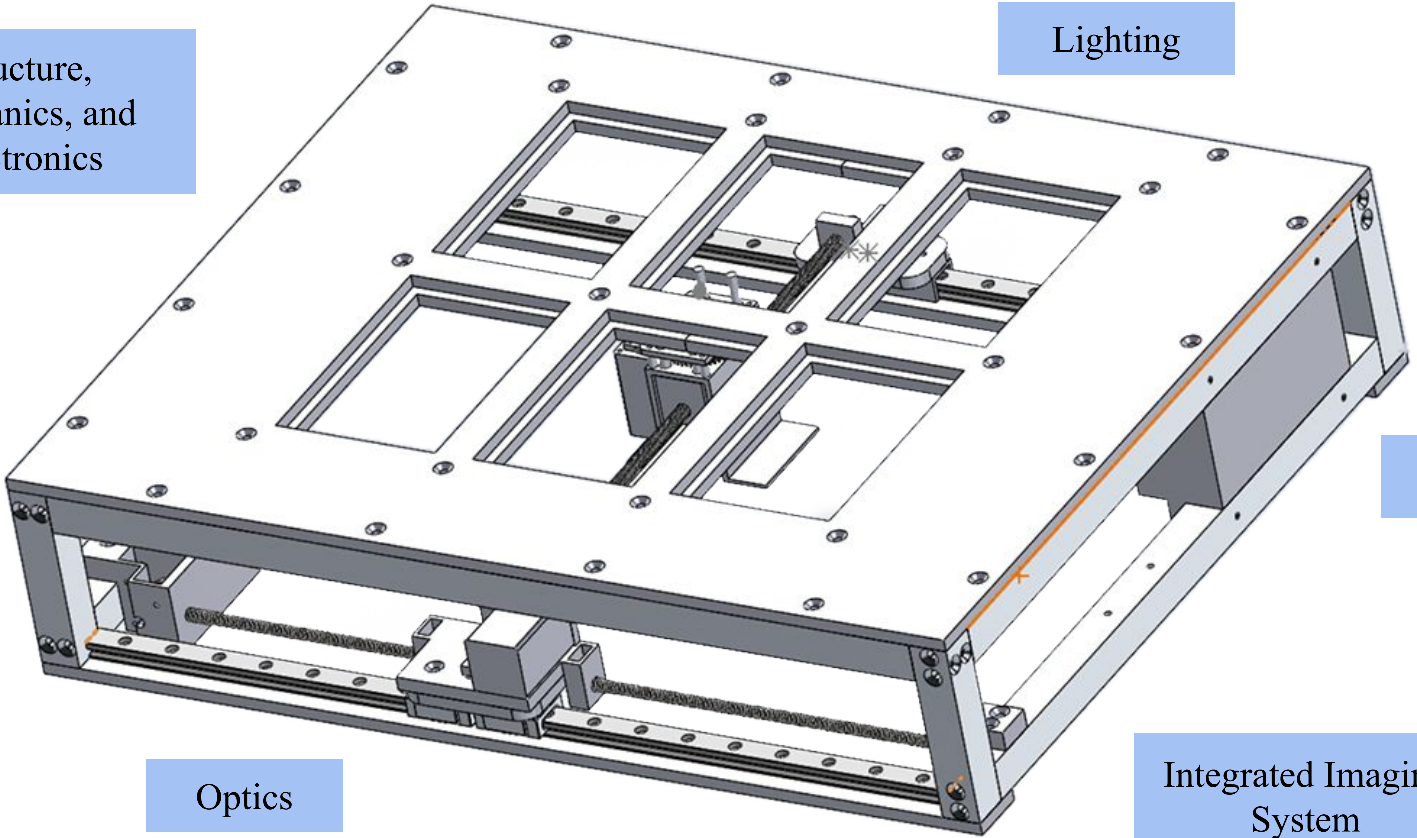
Technical Specifications

Imaging	• Generates research-grade image at 10x magnification • Camera components off-the-shelf or 3D printed
Data Storage	• Can locally store 10 GB of images • Sends images in real time
Resistance to Incubator Conditions	• 80%-100% Humidity • 5% CO_2 • 35-40 degrees (to accommodate 37-38 degrees)
Cost	• Does not involve long term/continuous cost for the client • Under $2000 in total production cost

CytoScope CAD Design

Structure, Mechanics, and Electronics

Lighting

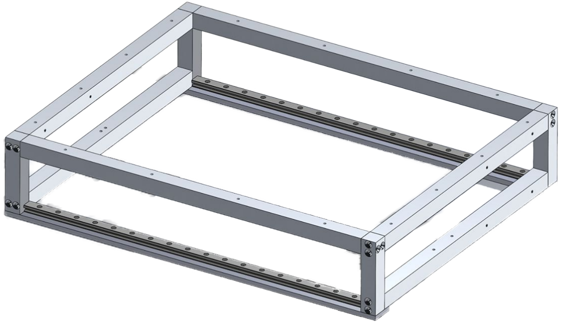


Code

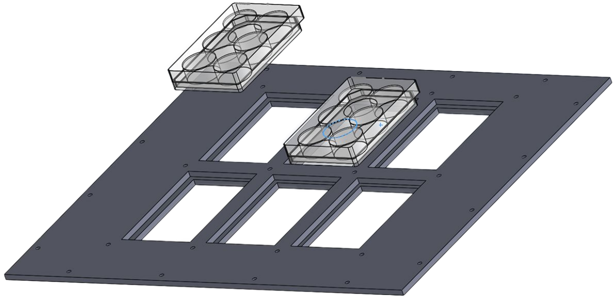
Optics

Integrated Imaging System

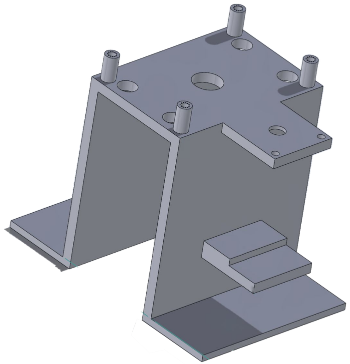
Structure



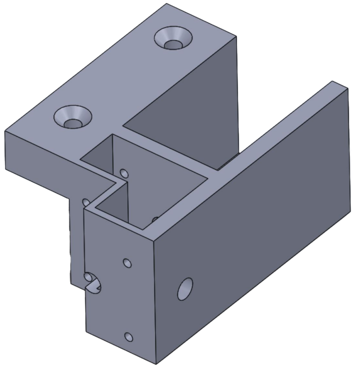
Aluminum frame & rail supports



Delrin cell plate holder



Objective Holder (PLA)



X axis motor mount (PLA)

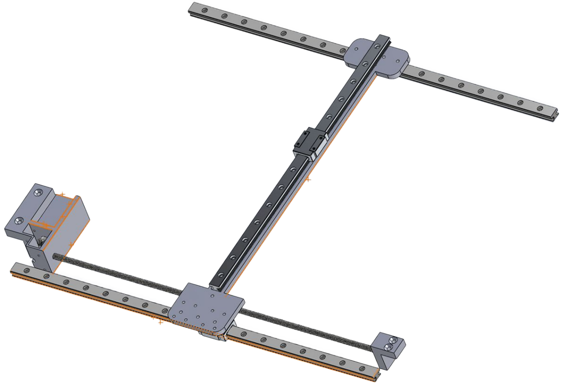
Specifications

✔ Size
- 50x40x20 cm

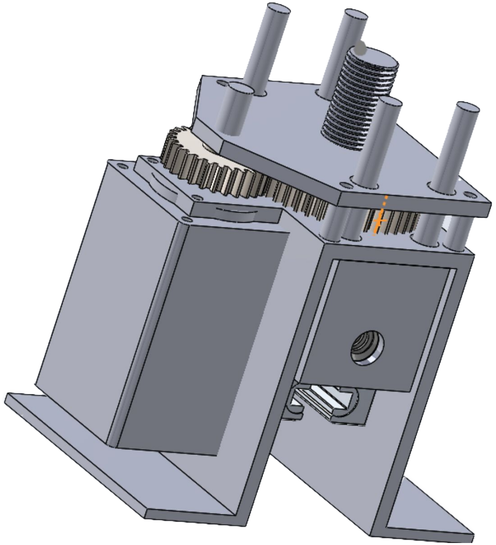
✔ Compatible with cell culture standard protocols
- Delrin + Aluminum

✔ Maintenance and calibration
- Ethanol-safe
- 3D-printed
- No need for maintenance

Mechanics



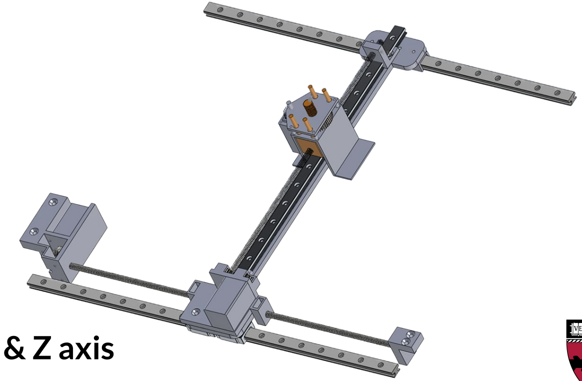
X & Y axis



Z axis

Specifications

- ✔ Size
  - 50x40x20 cm
- ✔ Compatible with cell culture standard protocols
  - Delrin + Aluminum
- ✔ Maintenance and calibration
  - Ethanol-safe
  - 3D-printed
  - No need for maintenance

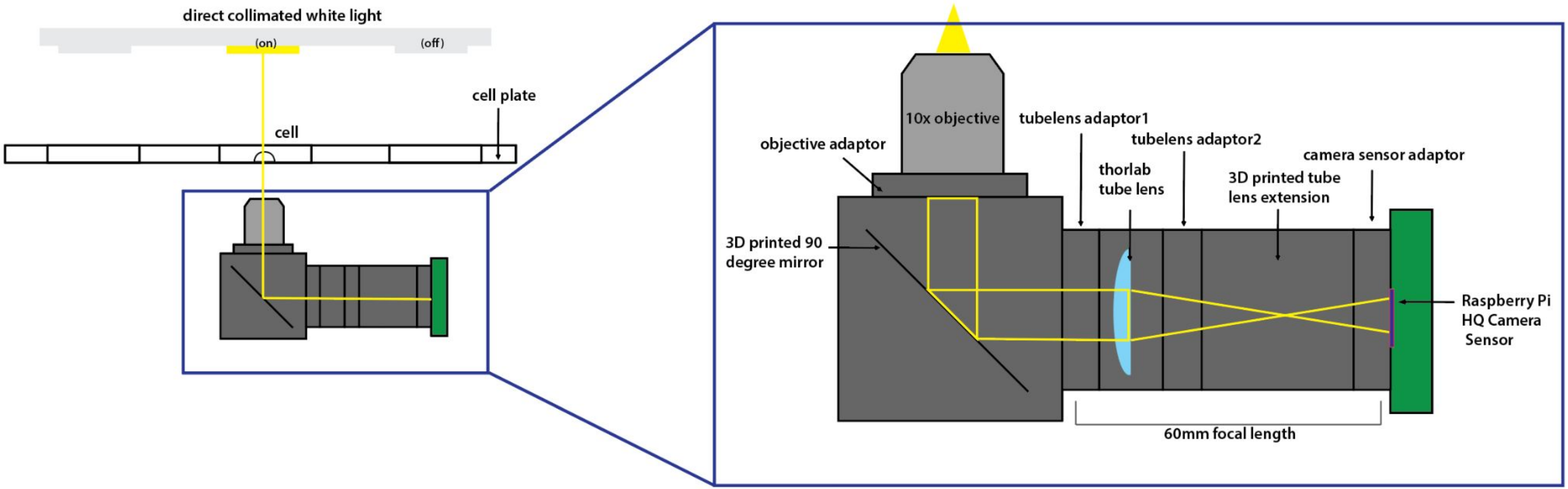


Complete X, Y, & Z axis

Electronics

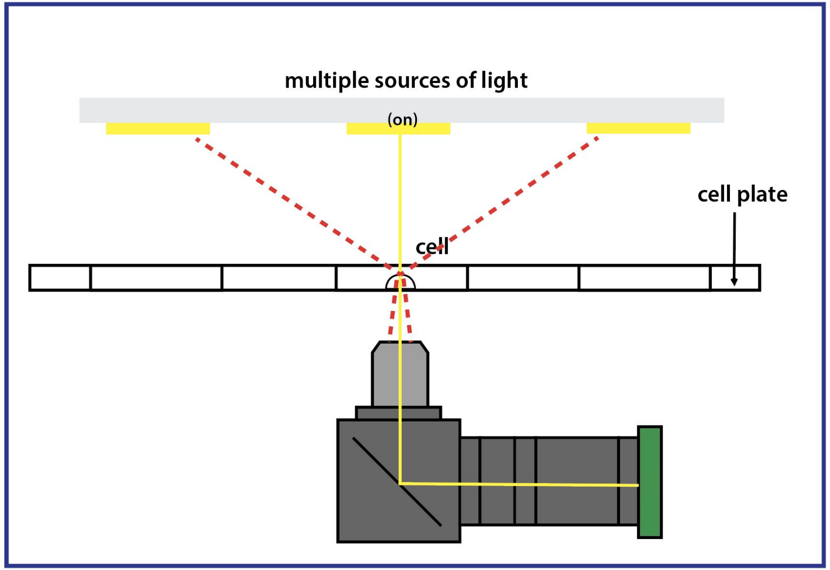
- Controlled by a Raspberry Pi 4 computer
- Motors controlled using add-on circuit boards
- Powered by 2 AC adapters
  - Raspberry Pi 4
    - Requires a 5V/3A adapter
  - Motors
    - Motor controller operates at 5V
    - Each motor draws 0.60-0.67A
    - 5V/4A adapter
- Lighting controlled using Arduino + 5V/3A adapter

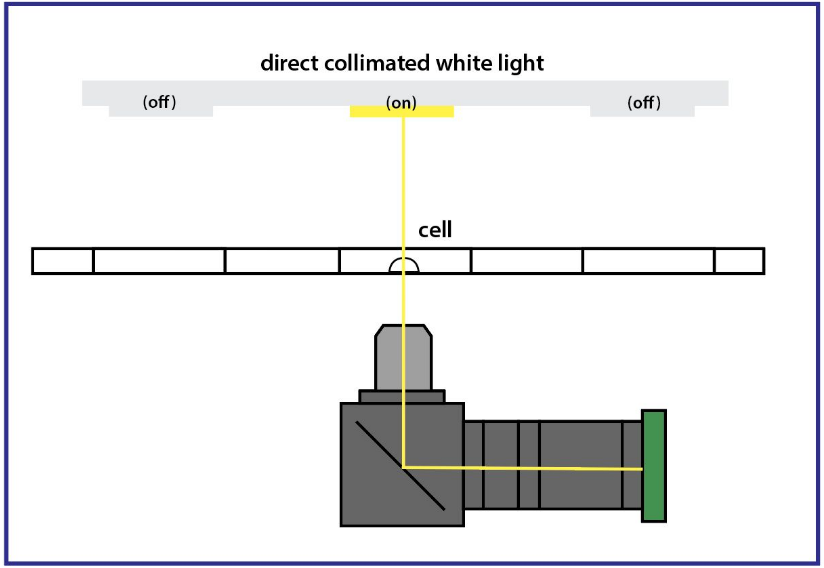
Optical System



A compact, light weight, high resolution brightfield microscope system that can be integrated onto a x-y-z actuating cart

Lighting

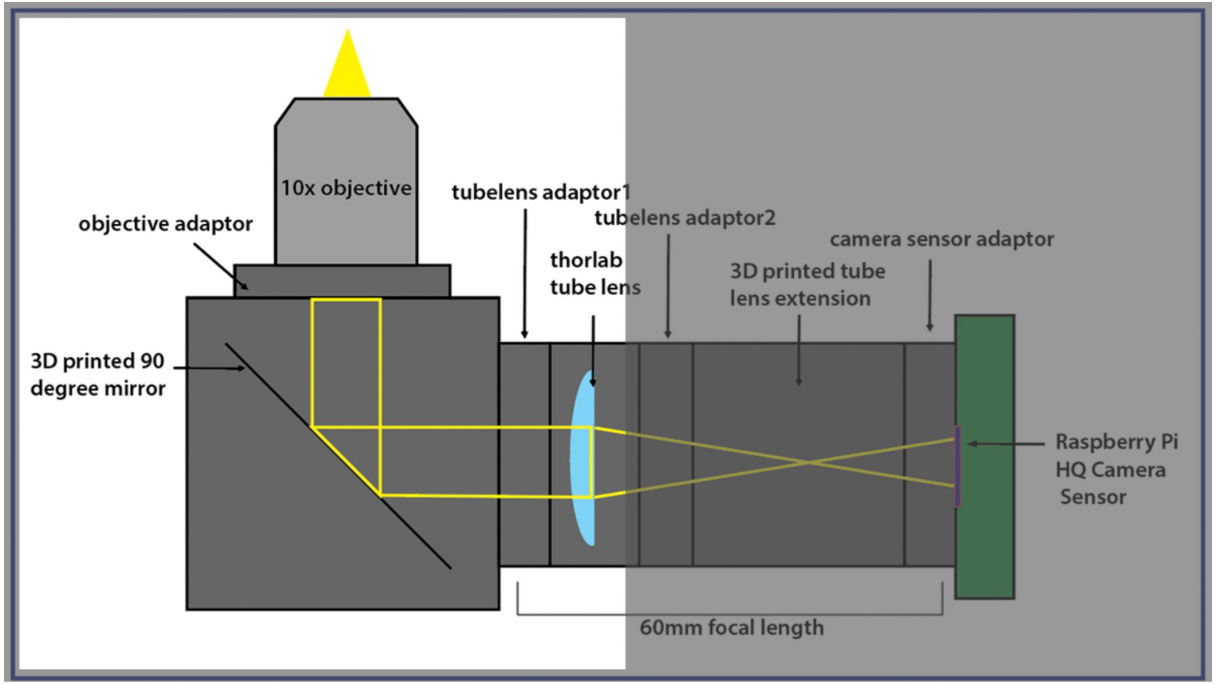




To maximize image quality, we needed to minimize the number of light sources and image cells under brightfield.

Chibitronics Light

Infinity-Corrected Objective



Function:

- An objective that creates an image at infinite focus

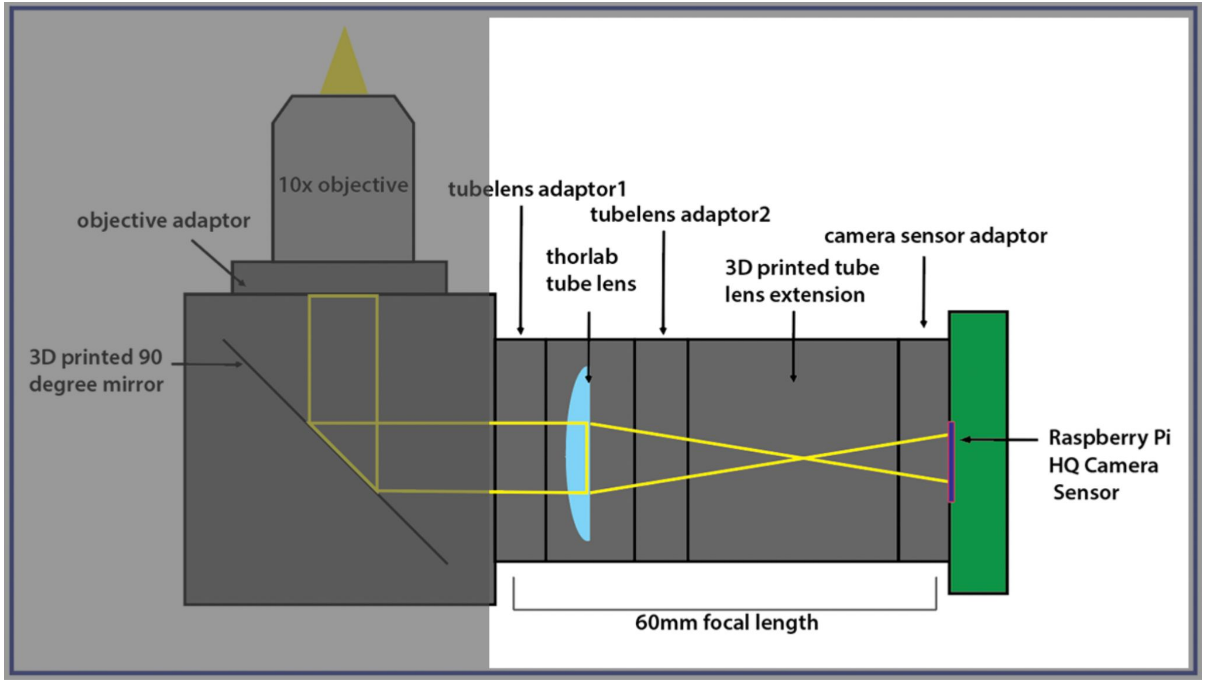
Constraints:

- Must use **a tube lens to focus the image**

Boli Optics 10x PH Contrast

The use of an infinitely focused objective provides the system with greater flexibility in terms of focal distance, enabling us to have a compact microscope design.

Tube Lens



Function:

- Focuses the image onto sensor

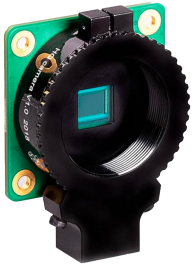
Constraints:

- Distance between tube lens and sensor **must be calculated**

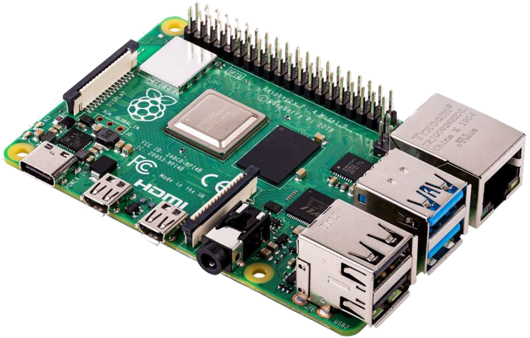
The tube lens focuses and amplifies the image from the infinitely focused objective onto the camera sensor.

Raspberry Pi Ecosystem

Using a Raspberry Pi camera allows for remote access and well-documented control/programming of the system.



Raspberry Pi HQ Camera



Raspberry Pi 4

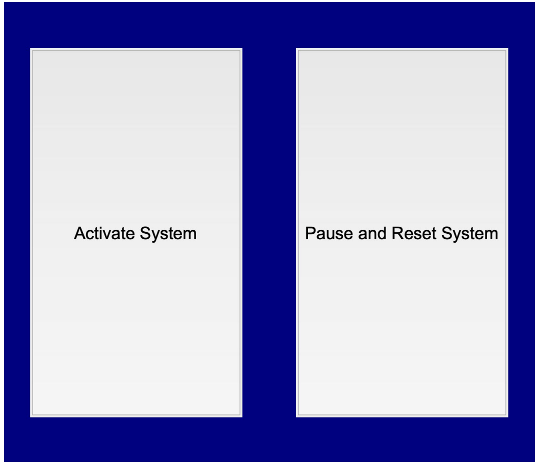
Integration with Python

Our computing and sensor components ensure that our system <u>meets our specifications of data storage, user interface, and imaging components.</u>

Control System and User Interface

Phase 1

User opens the Jupyter Notebook and an interface will pop up for the user to activate the system.



Phase 2

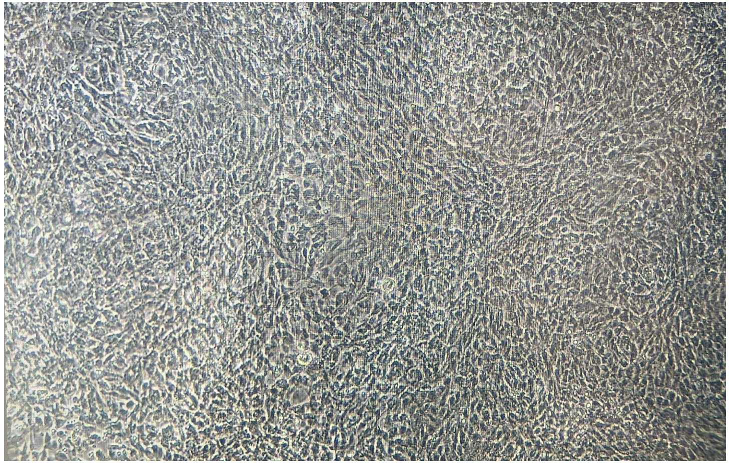
After activation, the system will prompt the user to input the name of the folder they want the images to appear in.

Phase 3

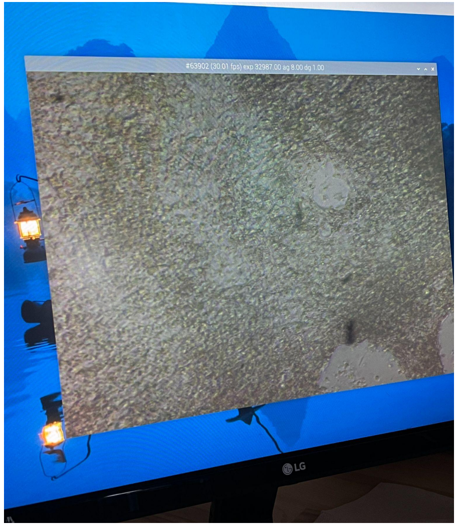
After input, the system will complete the imaging procedure across all six cell plates and six cell wells of each plate, named according to the cell plate, cell well, and time image was taken.

Microscope Images

Sample: Fixed Cardiomyocytes



Our System Images



Our system achieves similar imaging quality to a commercial microscope at 10x magnification.

Product Demonstration

Future Improvements

- Replace PLA with non degradable polymer
- Update code to allow for user selection of plate location and type
- Compatibility with different plate sizes (96- and 24-well plates)
- Integration of machine learning image classifier (automatically extract features)

CytoScope

Ultimately, *for only $1245.50*, and with only *three* total clicks and *one* keyboard input from the user, CytoScope is automatically able to image six cell plates without removing the samples from the incubator.